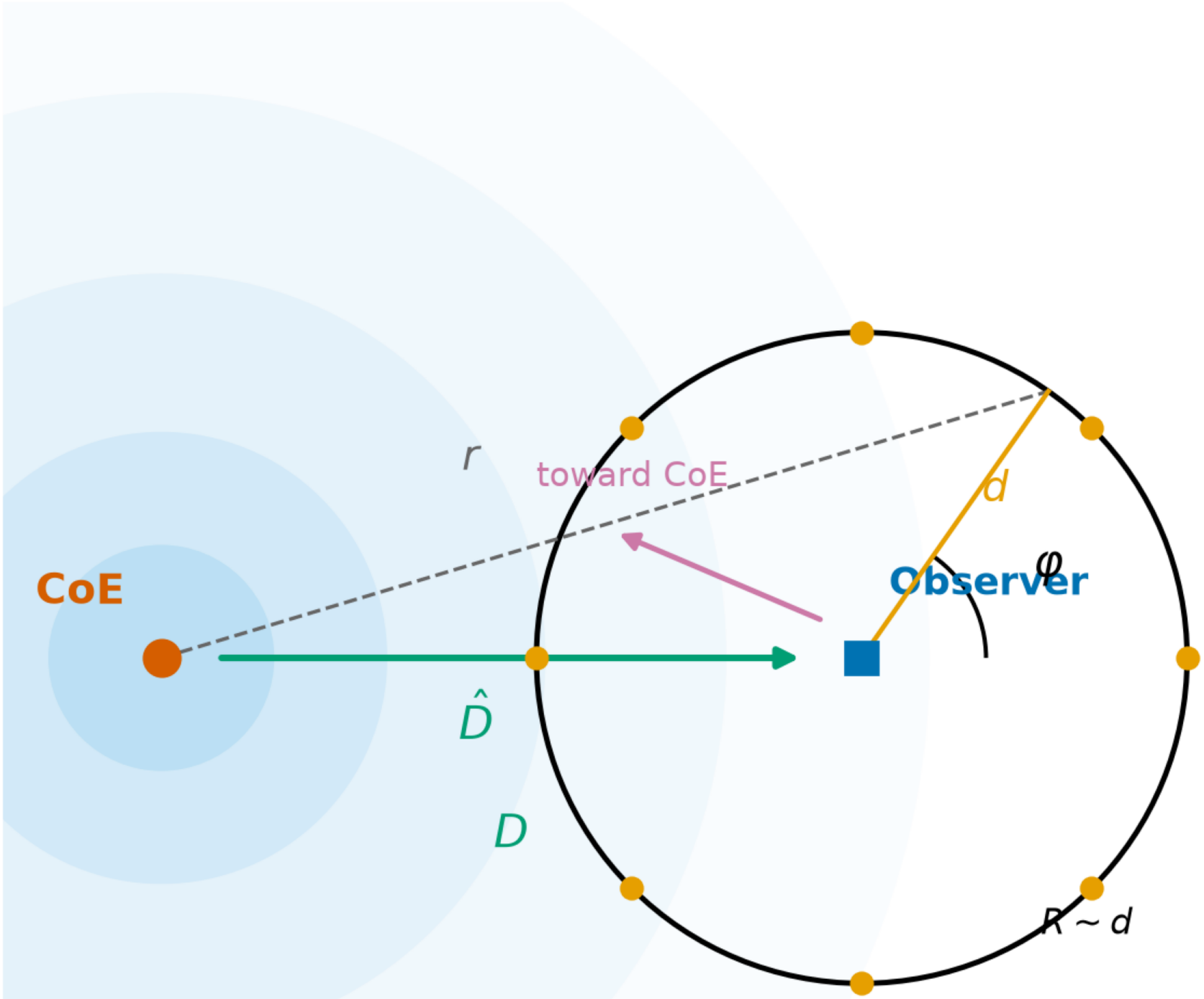
Geometry: CoE at origin, observer at $(D, 0)$, sources on circle about observer



Phase A eq. (3.13): exponential decreasing density & enclosed mass

$\rho(r)/\rho_0 = e^{-r/r_s}$

$m(r)/M = 1 - e^{-X}(1 + X + X^2/2)$

$r = D$ example ($D/r_s = 3$)

$\rho(r)/\rho_0$

$m(r)/M$

r/r_s

Angular match: first-principles model vs Migkas $A(\ell)$ — no dipole template

(a) $M = 0$ homology $\Rightarrow A/A_{\rm all} = 1$
(b) model unsmoothed ($b = 0$ cut)
(c) model $\times \Delta\ell = 90^\circ$ bin (Fig.6-like) ($\kappa = 0.15$, axis $\ell = 280^\circ$)
Migkas+2020 Fig.6 $A/A_{\rm all}$ (digitised)

$A/A_{\rm all}$

$B/B_{\rm all} = 1$ (isotropic slope)
Migkas+2020 Fig.6 $B/B_{\rm all}$ (digitised)

Model does not predict $B(\ell)$; panel shows Migkas slope isotropy check only.

$B/B_{\rm all}$

Galactic longitude ℓ (deg)

Model $H_{\rm obs}(\varphi)$ vs Migkas (from A) — qualitative comparison

(a) homology: $H_{\rm obs}/H - 1 = 0$

(b) model P_2-like ($\kappa = 0.15$), φ from $\ell = 280^\circ$

Migkas Fig.6 $\delta H/H$ from digitised A ($A \propto H^{-2}$)

Honest: Migkas $A(\ell)$ uses $\Delta\ell = 90^\circ$ bins (smoothed);
model is unsmoothed $b = 0$ cut. Shape mismatch listed as problems to fix.

$H_{\rm obs}/H - 1$

0.3

0.2

0.1

0.0

−0.1

−180

−90

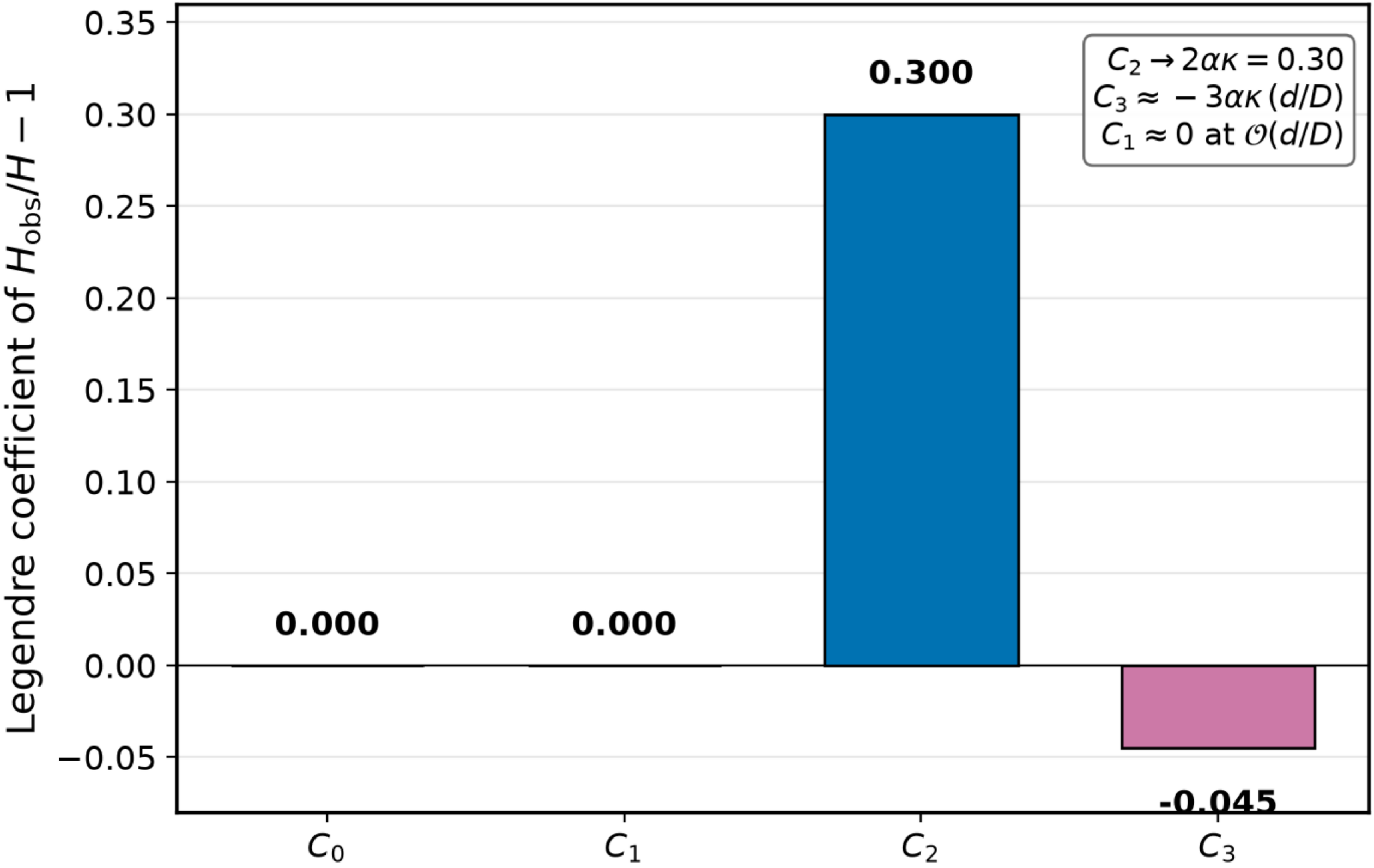
0

90

180

angle from preferred axis $\varphi = \ell - 280^\circ$ (deg, wrapped)

Full-sky multipoles at $d/D = 0.1$, $\kappa = 0.15$, $\alpha = 1.0$
C_2 dominant; $C_1 \approx 0$ (not a Migkas dipole)



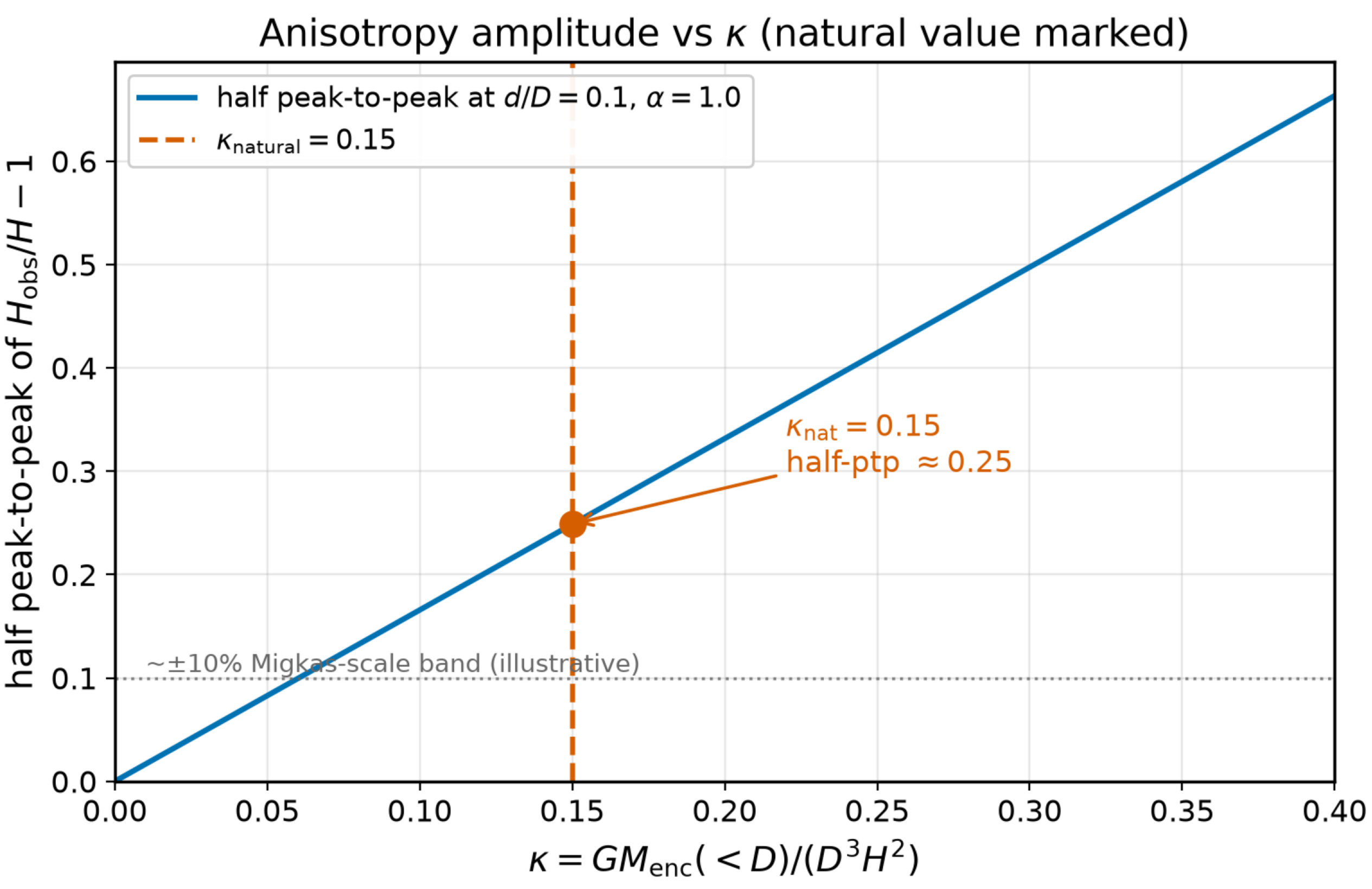
Sky map of leading P_2 anisotropy (Mollweide)

75°
60°
45°
30°
15°
0°
-15°
-30°
-45°
-60°
-75°
-150° -120° -90° -60° -30° 0° 30° 60° 90° 120° 150°

⋆ preferred axis $\hat{D}$ (CoE direction)
quadrupole lobes along $\pm\hat{D}$; not a dipole

−0.3 −0.2 −0.1 0.0 0.1 0.2 0.3

$H_{\rm obs}/H - 1$ (leading C_2P_2, $\kappa_{\rm nat} = 0.15$)


Anisotropy amplitude vs κ (natural value marked)
half peak-to-peak at d/D = 0.1, α = 1.0
κ_natural = 0.15
κ_nat = 0.15
half-ptp ≈ 0.25
~±10% Migkas-scale band (illustrative)
half peak-to-peak of H_obs/H − 1
κ = GM_enc(< D)/(D³H²)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40